TELIPORT
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Season 4

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Case Study No. 3 – Adaptive QoS Engine (PulseQoS)

Team Details

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THINK. BUILD. CREATE. WIN

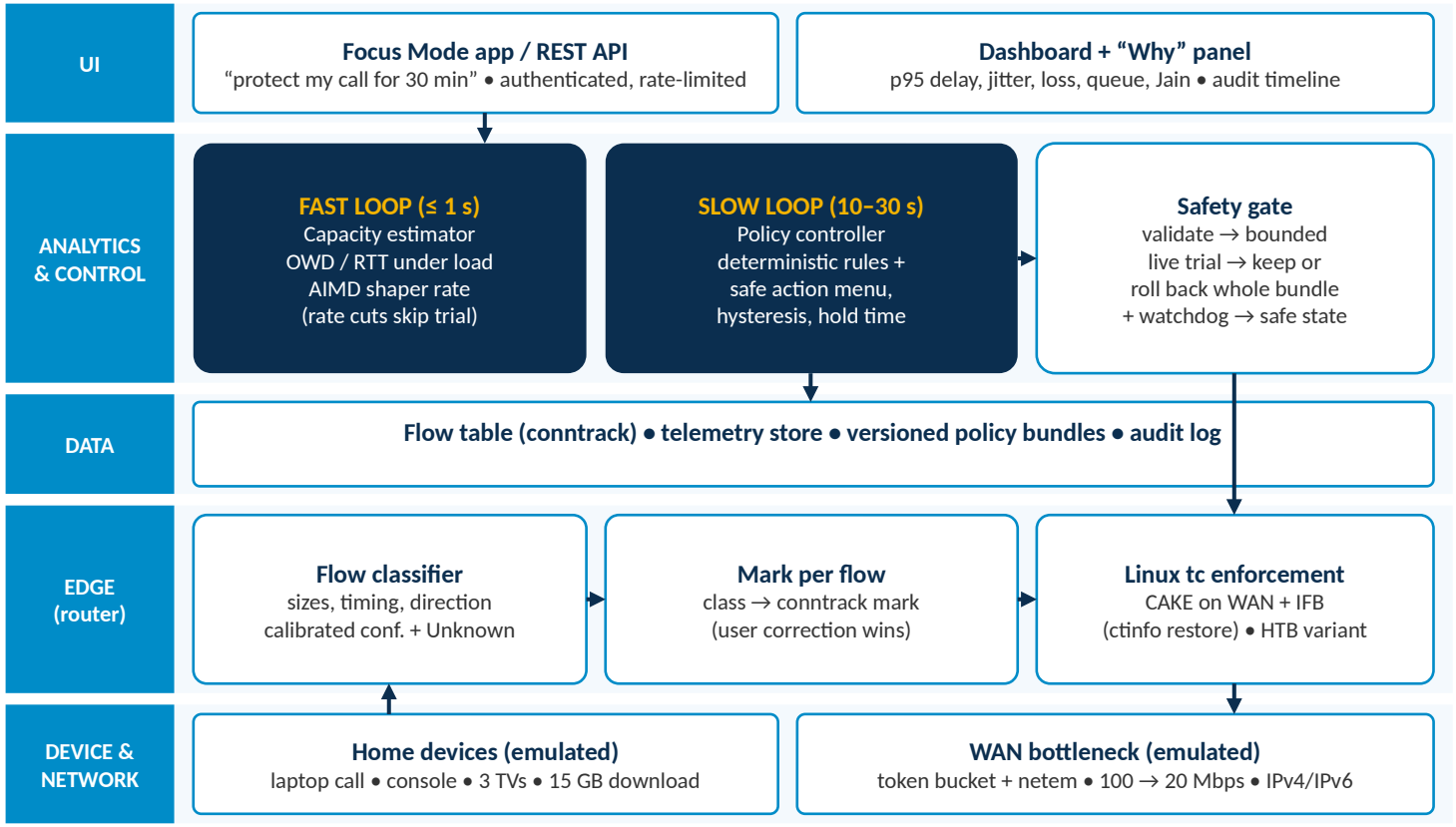
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PulseQoS: the call never notices the download

Intent-aware, self-verifying QoS controller for home broadband — payload-blind, built on open Linux tc

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ARCHITECTURE — one closed loop, two speeds



DESIGN RULES THAT KEEP IT CORRECT

- ▸ **Tins:** call audio + gaming → Voice; call video → Video (RFC 8837 WebRTC marking); bulk → Bulk; Unknown never boosted
- ▸ **Trust:** inbound DSCP from the Internet is ignored; download class is restored from our own conntrack mark
- ▸ **Two speeds:** capacity cuts act within a second; policy changes wait for the safety gate, so a link drop is never blocked by anti-flap rules
- ▸ **Privacy:** metadata only, no payload or domain logging; per-device views only to the account owner

PILOT EVIDENCE — measured, not assumed

p95 call-probe delay during a 4-stream download (ms)



Same download goodput in both: 18.7 Mbps of 20 — the call is protected without starving bulk.
Setup: Linux namespaces, 20 Mbit bottleneck, 50 pkt/s probes, 5 paired runs (medians; FIFO range 128–152 ms). Emulation; no WAN delay or Wi-Fi; ideal classification. Round 2 repeats this with CAKE, netem and all baselines.

UNDERSTAND → DECIDE → PROTECT → PROVE

- ▸ **Understand:** LightGBM on flow metadata, trained on real app captures, split by session; reports confidence, abstains to Unknown; metrics: macro-F1, wrong-boost rate, time-to-classify.
- ▸ **Decide:** Focus Mode with auto-expiry, priority ceiling and a daily Fairness Budget; a learned tuner stays only if it beats the rule controller in A/B runs.
- ▸ **Protect:** fast-abort on call latency, whole-bundle rollback with read-back check, watchdog → known safe state.
- ▸ **Prove:** one command runs 9 scenarios × 6 systems (FIFO, CAKE, CAKE+autorate, rule QoS, ours, ours+ML), 10 runs each: p95/p99 delay, jitter, loss, RPM, Jain, settle time, CPU.

Boundary: reference implementation in emulation, labelled as such. A product also needs router hardware, flow-offload handling, gigabit CPU budget, Wi-Fi airtime fairness and a security review of the intent API.